Linux Filesystem

Exercise 1 - Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   40G  0 disk
├─sda1   8:1    0 37.9G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  2.1G  0 part [SWAP]
sr0     11:0    1 1024M  1 rom
```

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4   1.0         17583018-79f2-4ff6-a7f6-1f8949505937   17.6G    47% /
├─sda2
└─sda5 swap   1           69f05d61-a7d9-4638-9d17-32ac3bb3c606                [SWAP]
sr0
```

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ df -T
Filesystem     Type      1K-blocks      Used Available Use% Mounted on
udev           devtmpfs    1969876         0   1969876   0% /dev
tmpfs          tmpfs        401136      1008    400128   1% /run
/dev/sda1      ext4       38818336  18310976  18503292  50% /
tmpfs          tmpfs       2005680         4   2005676   1% /dev/shm
none           tmpfs          1024         0      1024   0% /run/credentials/systemd-journald.service
tmpfs          tmpfs       2005680         8   2005672   1% /tmp
Kali           vboxsf    211430396 197920552  13509844  94% /media/sf_Kali
none           tmpfs          1024         0      1024   0% /run/credentials/getty@tty1.service
tmpfs          tmpfs        401136       108    401028   1% /run/user/1000
```

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           392M 1008K  391M   1% /run
/dev/sda1        38G   18G   18G  50% /
tmpfs           2.0G  4.0K  2.0G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.0G  8.0K  2.0G   1% /tmp
Kali            202G  189G   13G  94% /media/sf_Kali
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           392M  108K  392M   1% /run/user/1000
```

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ df -T /
Filesystem     Type 1K-blocks     Used Available Use% Mounted on
/dev/sda1      ext4  38818336 18310976  18503292  50% /
```

Determine:

1. **The storage devices available.**

   The main storage device available is /dev/sda, which has a total size of 40 GB. The system also has /dev/sr0, a ROM device with a size of 1024 MB.

2. **The partitions available on each device.**

   The /dev/sda disk contains the following partitions:

   - /dev/sda1 – 37.9 GB
   - /dev/sda2 – 1 KB
   - /dev/sda5 – 2.1 GB

3. **The filesystem type used by each partition.**
   - /dev/sda1 uses the ext4 filesystem.
   - /dev/sda2 has no filesystem shown.
   - /dev/sda5 is used as swap.

4. **The device/partition on which `/` is located.**
   The root directory / is located on:

   /dev/sda1

5. **The filesystem type used for `/`.**

   ext4

6. **The total, used, and available storage space.**
   Total space = 38GB
   Used space = 18GB
   Available space = 18GB

Suggested commands:

lsblk

lsblk -f

df -T

df -h

Record your findings in a table.

Device/Partition	Filesystem Type	Size	Mount Point
/dev/sda	Disk	40GB	-
/dev/sda1	ext4	37.9GB	/
/dev/sda2	No filesystem	1KB	-
/dev/sda5	swap	2.1GB	[SWAP]
/dev/sr0	ROM	1024MB	-

Exercise 2 - Disk → Partition → Filesystem → Mount Point

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system. Use the actual values observed on your machine.

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4   1.0         17583018-79f2-4ff6-a7f6-1f8949505937   17.6G    47% /
├─sda2
└─sda5 swap   1           69f05d61-a7d9-4638-9d17-32ac3bb3c606                [SWAP]
sr0
```

```
/dev/sda (40 GB Disk)
|
├── /dev/sda1 (37.9 GB Partition)
|      |
|      └── ext4 Filesystem
|              |
|              └── / (Root Mount Point)
|
├── /dev/sda2 (1 KB Partition)
|      |
|      └── No filesystem shown
|
└── /dev/sda5 (2.1 GB Partition)
       |
       └── swap Filesystem
               |
               └── [SWAP]
```

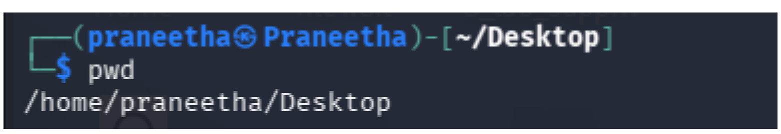
Explain in 2–3 sentences the relationship between these four components.

The disk /dev/sda is the main storage device and is divided into partitions such as /dev/sda1, /dev/sda2, and /dev/sda5. The /dev/sda1 partition contains an ext4 filesystem and is mounted at /, which is the root of the Linux filesystem, while /dev/sda5 is used as swap space.

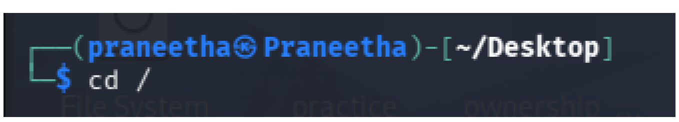
Exercise 3 - Explore the Root Filesystem

Start from your current location.

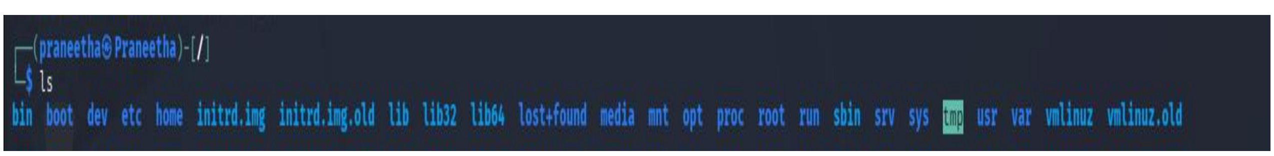
1. Determine your current working directory.



2. Navigate to the root directory.



3. Display the contents of the root directory.



4. Identify all directories directly available under `/`.

```
┌──(praneetha㉿Praneetha)-[/]
└─$ ls -d */
bin/  boot/  dev/  etc/  home/  lib/  lib32/  lib64/  lost+found/  media/  mnt/  opt/  proc/  root/  run/  sbin/  srv/  sys/  tmp/  usr/  var/
```

5. Determine whether hidden entries are present.

```
┌──(praneetha㉿Praneetha)-[/]
└─$ ls -la
total 76
drwxr-xr-x  19 root root  4096 Aug 11 16:57 .
drwxr-xr-x  19 root root  4096 Aug 11 16:57 ..
lrwxrwxrwx   1 root root     7 Jul 30 23:28 bin → usr/bin
drwxr-xr-x   3 root root  4096 Aug 11 17:00 boot
drwx──────   2 root root  4096 Jul 30 23:50 .cache
drwxr-xr-x  18 root root  3280 Sep  6 08:33 dev
drwxr-xr-x 185 root root 12288 Sep  6 08:33 etc
drwxr-xr-x   6 root root  4096 Aug 31 21:41 home
lrwxrwxrwx   1 root root    33 Aug 11 16:57 initrd.img → boot/initrd.img-7.0.12+kali-amd64
lrwxrwxrwx   1 root root    34 Jul 30 23:30 initrd.img.old → boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx   1 root root     7 Jul 30 23:28 lib → usr/lib
lrwxrwxrwx   1 root root     9 Jul 30 23:40 lib32 → usr/lib32
lrwxrwxrwx   1 root root     9 Jul 30 23:28 lib64 → usr/lib64
drwx──────   2 root root 16384 Jul 30 23:28 lost+found
drwxr-xr-x   4 root root  4096 Aug 13 18:29 media
drwxr-xr-x   2 root root  4096 Jul 30 23:29 mnt
drwxr-xr-x   3 root root  4096 Jul 30 23:40 opt
dr-xr-xr-x 241 root root     0 Sep  6 08:32 proc
drwx──────   4 root root  4096 Sep  6 08:33 root
drwxr-xr-x  40 root root   940 Sep  6 08:33 run
lrwxrwxrwx   1 root root     8 Jul 30 23:28 sbin → usr/sbin
drwxr-xr-x   3 root root  4096 Jul 30 23:42 srv
dr-xr-xr-x  13 root root     0 Sep  6 08:32 sys
drwxrwxrwt  11 root root   300 Sep  6 09:01 tmp
drwxr-xr-x  15 root root  4096 Jul 30 23:40 usr
drwxr-xr-x  12 root root  4096 Jul 30 23:57 var
lrwxrwxrwx   1 root root    30 Aug 11 16:57 vmlinuz → boot/vmlinuz-7.0.12+kali-amd64
lrwxrwxrwx   1 root root    31 Jul 30 23:30 vmlinuz.old → boot/vmlinuz-6.19.14+kali-amd64
```

Linux directories whose names begin with . are hidden from a normal ls command.

Exercise 4 - Linux Filesystem Hierarchy Investigation

Investigate the following locations:

/home, /root,

/etc, /var,

/var/log, /tmp,

/usr, /boot,

/dev, /proc

Commands used:

ls -la /home

ls -la /root

ls -la /etc

ls -la /var

ls -la /var/log

ls -la /tmp

ls -la /usr

ls -la /boot

ls -la /dev

ls -la /proc

For each location:

1. Examine its contents.

2. Identify at least three entries, where available.

3. Determine the likely purpose of the directory.

Directory	Example Contents	Purpose
/home	alice, bob, charlie, praneetha	Contains the personal home directories of users
/root	Access denied	Home directory of the root user
/etc	passwd, group, hostname	Contains system-wide configuration files
/var	cache, lib, log, spool	Stores variable data generated during system operation
/var/log	boot.log, dpkg.log, wtmp, Xorg.0.log	Stores system and application log files
/tmp	.font-unix, .ICE-unix, systemd-private-*	Stores temporary files and directories
/usr	bin, lib, share, include	Contains user programs, libraries and shared resources
/boot	grub, vmlinuz-*, initrd.img-*	Contains files required for system booting
/dev	null, zero, sda, sda1, sr0	Contains device files and device interfaces
/proc	Process and system information	Provides dynamic information about the kernel and running processes

---

Exercise 5 - Find Your Home Directory

Without assuming its location:

1. Determine your username - whoami

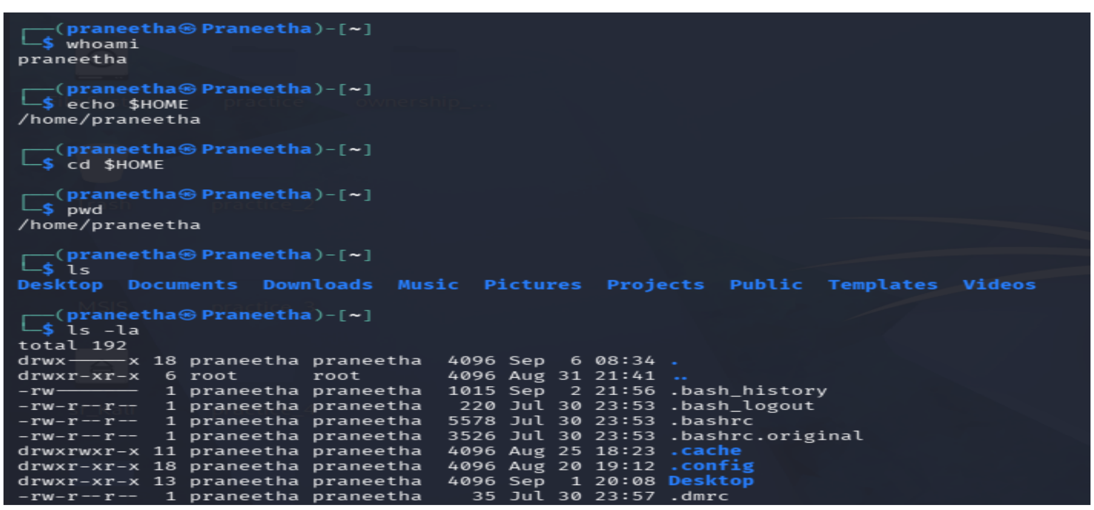
2. Determine your home directory - echo $HOME

3. Navigate to your home directory - cd $HOME

4. Display its absolute path - pwd

5. List its normal contents - ls

6. List all contents, including hidden files - ls -la



Exercise 6 - Hidden Files Investigation

Examine your home directory.

Compare:

ls

ls -a

- ls command displays only the normal files and directories
- ls -a command additionally displays hidden entries such as .bashrc, .bash_history, .config, .ssh, .cache, and .zshrc.

Identify at least 2 hidden files/directories.

Hidden file: .bashrc, .bash_history

Hidden directory: .config, .ssh

Answer:

1. How does Linux identify a hidden file?

Linux identifies a hidden file or directory by a dot (.) at the beginning of its name.

2. Why does a normal `ls` not display these files?

The normal ls command does not display hidden files by default. They can be displayed using

ls -a

3. What do '.' and '..' mean?

'.' represents the current directory.

'..' represents the parent directory.

4. Can a hidden file still be accessed if its name is known?

Yes. A hidden file is not protected simply because it is hidden. If its name and location are known and the user has sufficient permissions, it can be accessed normally.

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden files are not automatically safe or malicious. Many legitimate Linux configuration files are hidden, such as .bashrc and .config. However, malicious users or programs can also hide files to avoid casual detection, so hidden files should be investigated based on their purpose, contents, ownership, permissions, and timestamps.

Exercise 7 - Build a Small Filesystem Hierarchy

Explore: mkdir

Create the following files:

cyber_lab/evidence/network/traffic.txt

cyber_lab/evidence/system/processes.txt

cyber_lab/logs/security.log

cyber_lab/scripts/check.sh

cyber_lab/reports/investigation.txt

Use appropriate Linux commands to create the structure.

```
┌──(praneetha㉿Praneetha)-[~]
└─$ cd ~

┌──(praneetha㉿Praneetha)-[~]
└─$ mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports

┌──(praneetha㉿Praneetha)-[~]
└─$ touch cyber_lab/evidence/network/traffic.txt
touch cyber_lab/evidence/system/processes.txt
touch cyber_lab/logs/security.log
touch cyber_lab/scripts/check.sh
touch cyber_lab/reports/investigation.txt
```

Verify that the required hierarchy has been created successfully.

Hint: tree

```
┌──(praneetha㉿Praneetha)-[~]
└─$ tree cyber_lab
cyber_lab
├── evidence
│   ├── network
│   │   └── traffic.txt
│   └── system
│       └── processes.txt
├── logs
│   └── security.log
├── reports
│   └── investigation.txt
└── scripts
    └── check.sh

7 directories, 5 files
```

Exercise 8 - Absolute and Relative Paths

Assume your current directory is:

~/cyber_lab

```
┌──(praneetha㉿Praneetha)-[~]
└─$ cd ~/cyber_lab

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ pwd
/home/praneetha/cyber_lab
```

Perform the following:

1. Navigate to `evidence/network` using a relative path.

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ cd evidence/network

┌──(praneetha㉿Praneetha)-[~/cyber_lab/evidence/network]
└─$ pwd
/home/praneetha/cyber_lab/evidence/network
```

2. Navigate to `/etc` using an absolute path.

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/evidence/network]
└─$ cd /etc

┌──(praneetha㉿Praneetha)-[/etc]
└─$ pwd
/etc
```

3. Return to your home directory using the shortest possible command.

```
┌──(praneetha㉿Praneetha)-[/etc]
└─$ cd ~
```

4. Navigate to `cyber_lab/reports`.

```
┌──(praneetha㉿Praneetha)-[~]
└─$ cd ~/cyber_lab/reports

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ pwd
/home/praneetha/cyber_lab/reports
```

5. Move one directory upward.

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ cd ..

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ pwd
/home/praneetha/cyber_lab
```

6. Display the current directory after every operation.

\

*** What is an absolute path?**

A path that starts from the root directory / and gives the complete location of a file or directory.

*** What is a relative path?**

A path interpreted from the current working directory.

*** What is the difference between '/' and '~'?**

'/' represents the root directory of the Linux filesystem.

'~' represents the current user's home directory.

Exercise 9 - Identify File Types

Investigate the following:

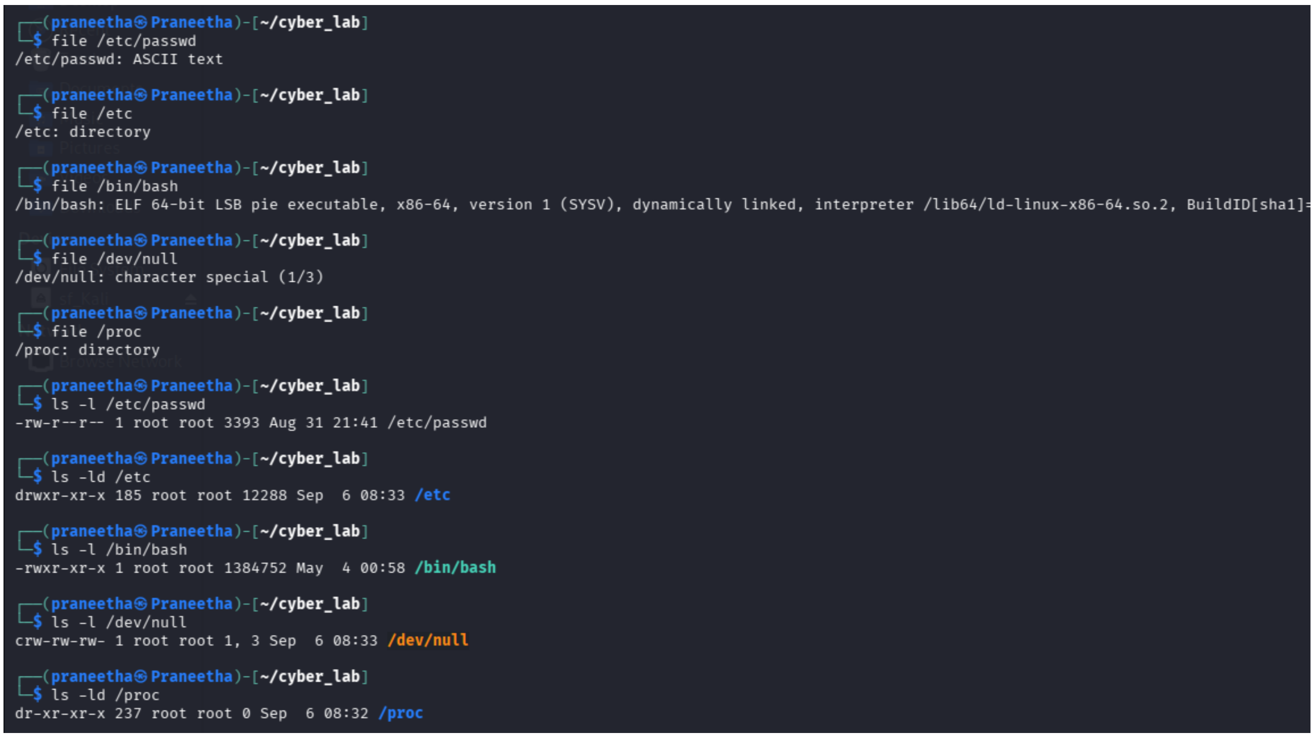
/etc/passwd

/etc

/bin/bash

/dev/null

/proc

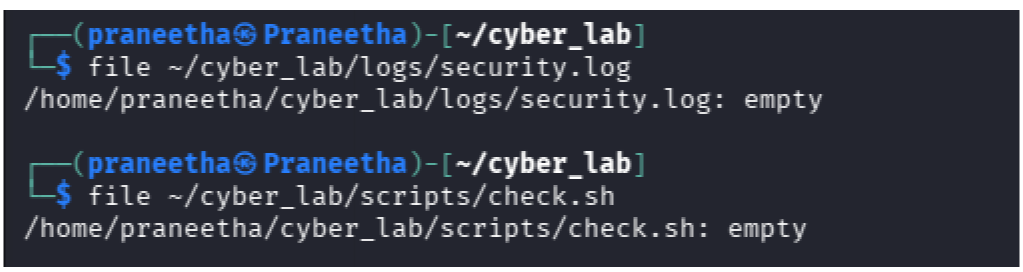


```
┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file /etc/passwd
/etc/passwd: ASCII text

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file /etc
/etc: directory

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file /bin/bash
/bin/bash: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file /dev/null
/dev/null: character special (1/3)

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file /proc
/proc: directory

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3393 Aug 31 21:41 /etc/passwd

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -ld /etc
drwxr-xr-x 185 root root 12288 Sep  6 08:33 /etc

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -l /bin/bash
-rwxr-xr-x 1 root root 1384752 May  4 00:58 /bin/bash

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Sep  6 08:33 /dev/null

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -ld /proc
dr-xr-xr-x 237 root root 0 Sep  6 08:32 /proc
```

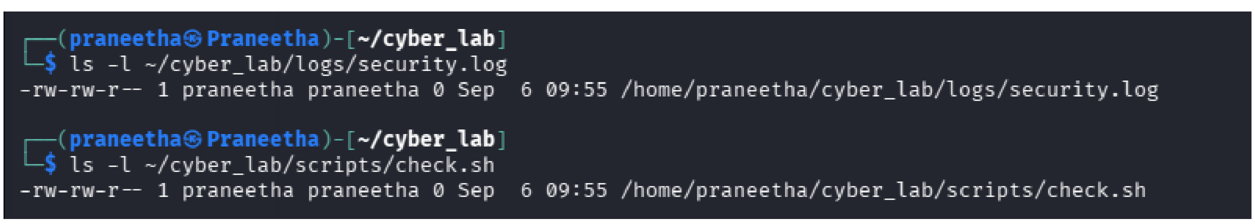
Also investigate the files created in Exercise 7.

file <filename>

ls -l <filename>



```
┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file ~/cyber_lab/logs/security.log
/home/praneetha/cyber_lab/logs/security.log: empty

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ file ~/cyber_lab/scripts/check.sh
/home/praneetha/cyber_lab/scripts/check.sh: empty
```



```
┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -l ~/cyber_lab/logs/security.log
-rw-rw-r-- 1 praneetha praneetha 0 Sep  6 09:55 /home/praneetha/cyber_lab/logs/security.log

┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ ls -l ~/cyber_lab/scripts/check.sh
-rw-rw-r-- 1 praneetha praneetha 0 Sep  6 09:55 /home/praneetha/cyber_lab/scripts/check.sh
```

Object	**Type**
/etc/passwd	ASCII text / Regular file
/etc	Directory
/bin/bash	ELF 64-bit executable
/dev/null	Character special device
/proc	Directory
security.log	Empty regular file
check.sh	Empty regular file

Exercise 10 - Examine File Metadata

Use:

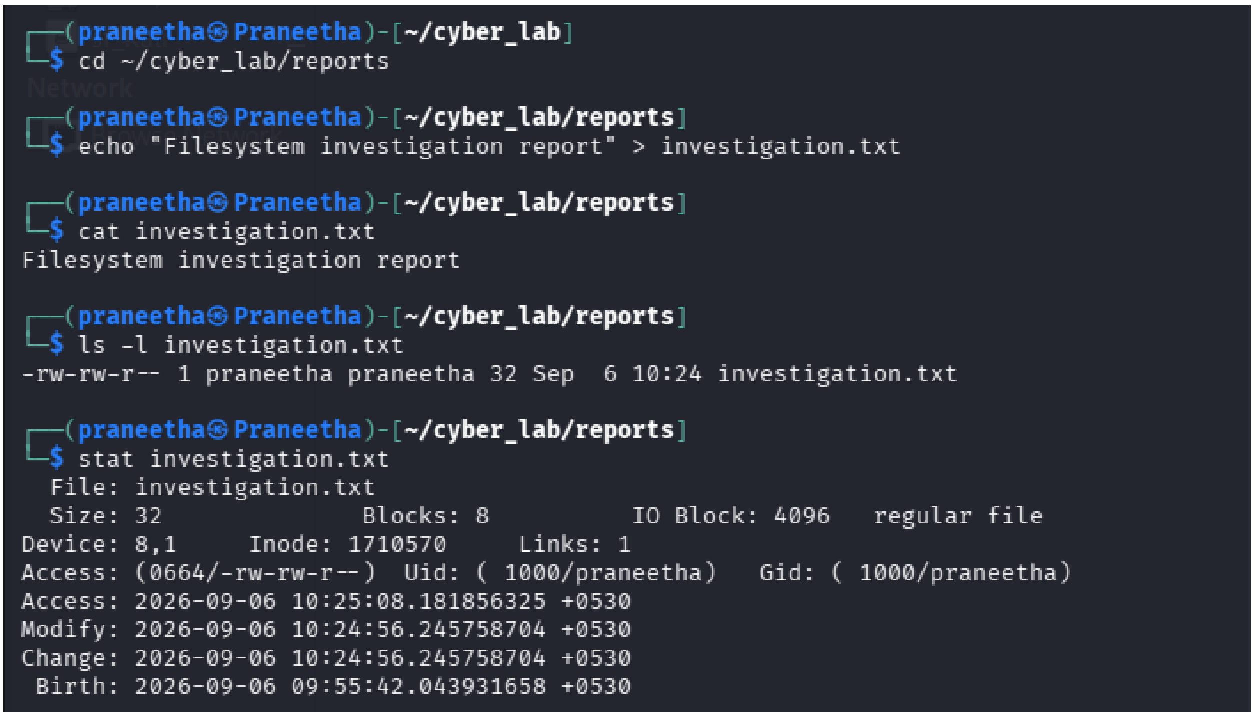
~/cyber_lab/reports/investigation.txt

Add a small amount of text to the file.

Examine it using:

ls -l

stat investigation.txt



```
┌──(praneetha㉿Praneetha)-[~/cyber_lab]
└─$ cd ~/cyber_lab/reports

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ echo "Filesystem investigation report" > investigation.txt

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ cat investigation.txt
Filesystem investigation report

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -l investigation.txt
-rw-rw-r-- 1 praneetha praneetha 32 Sep  6 10:24 investigation.txt

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ stat investigation.txt
  File: investigation.txt
  Size: 32              Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 1710570     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/praneetha)   Gid: ( 1000/praneetha)
Access: 2026-09-06 10:25:08.181856325 +0530
Modify: 2026-09-06 10:24:56.245758704 +0530
Change: 2026-09-06 10:24:56.245758704 +0530
 Birth: 2026-09-06 09:55:42.043931658 +0530
```

Identify:

* Filename: investigation.txt

* File size: 32 bytes

* File type: Regular file

* Owner: praneetha

* Group: praneetha

* Inode number: 1710570

* Access time: 2026-09-06 10:25:08 +0530

* Modification time: 2026-09-06 10:24:56 +0530

* Change time: 2026-09-06 10:24:56 +0530

Answer:

1. **What is metadata?**

Metadata is information that describes a file, such as its name, size, type, owner, permissions, inode number, and timestamps.

2. **Is metadata the same as file content?**

No

File content is the actual information stored inside the file.
Metadata describes the file and provides information about how the filesystem manages it.

For example, the content of investigation.txt is:

Filesystem investigation report

while its size, owner, permissions, inode number, and timestamps are metadata.

3. **Which timestamp changes when file contents are modified?**

mtime (modification time) changes directly. ctime also updates as a side effect (since the inode's size/mtime metadata changed), but atime only changes on read access, not on write.

Exercise 11 - Explore `/etc`

Investigate:

/etc

`` ``` ``

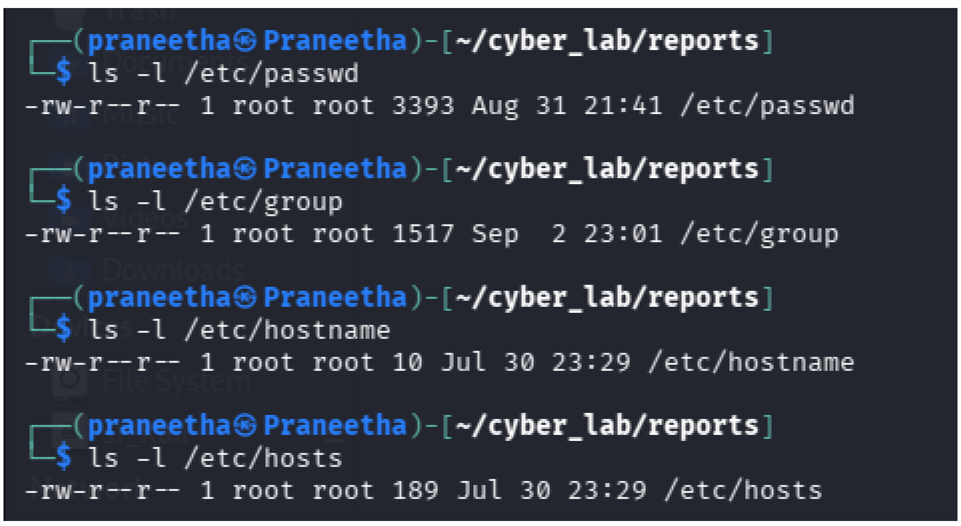
Locate:

passwd

group

hostname

hosts



```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3393 Aug 31 21:41 /etc/passwd

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -l /etc/group
-rw-r--r-- 1 root root 1517 Sep  2 23:01 /etc/group

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -l /etc/hostname
-rw-r--r-- 1 root root 10 Jul 30 23:29 /etc/hostname

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -l /etc/hosts
-rw-r--r-- 1 root root 189 Jul 30 23:29 /etc/hosts
```

Note: Without modifying any system files, examine their contents where permitted.

cat /etc/passwd

cat /etc/group

cat /etc/hostname

cat /etc/hosts

Answer:

1. **What kind of information is primarily stored under `/etc`?**

/etc primarily contains system-wide configuration files and configuration information.

2. What information does `/etc/passwd` contain?

It contains account information such as usernames, user IDs, group IDs, home directories, and login shells.

3. What information does `/etc/hostname` contain?

It contains the system's hostname.

4. Why is `/etc` security-sensitive?

It contains important system configuration and account-related information. Unauthorized modification of these files could affect authentication, networking, services, and system behavior.

Exercise 12 - Explore Linux Log Storage

Investigate:

/var/log

List available log files and directories

ls -la /var/log

```

Identify:

* Available log files

find /var/log -maxdepth 1 -type f -print

**Available log files:** boot.log, dpkg.log, wtmp, btmp, Xorg.0.log, alternatives.log, fontconfig.log, and archived log files.

* Available log directories

find /var/log -maxdepth 1 -type d -print

**Available log directories:** apache2, apt, journal, mysql, nginx, postgresql, redis, samba, gvm, and others.

* Recently modified entries

ls -lt /var/log

**Recently modified entries:** boot.log, boot.log.1, wtmp, wtmp.db, lightdm, and Xorg.0.log.

* Entries you can read

find /var/log -type f -readable -print

**Entries you can read:** Logs such as dpkg.log, Xorg.0.log, apache2/access.log, apache2/error.log, and nginx/access.log are readable.

* Entries you cannot read as a normal user

find /var/log ! -readable -print

**Entries you cannot read as a normal user:** Access was denied for directories such as private, mosquitto, inetsim, lightdm, and speech-dispatche
```

Answer:

1. Why does Linux maintain logs?

Linux maintains logs to record system, application, service, and user activities. Logs provide information about system events, errors, service operations, login activity, and other important events occurring on the system.

2. Why are logs useful during incident investigation?

Logs are useful because they provide evidence of activities that occurred on the system. During an incident investigation, they can help identify login activity, system events, application activity, errors, and other events that may help investigators understand what happened.

3. Why might access to some logs be restricted?

Access to some logs may be restricted because they can contain sensitive system or user information. Linux uses ownership and file permissions to prevent unauthorized users from reading or modifying protected log information.

Exercise 13 - Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Commands Used

Check the contents of /tmp

ls -la /tmp

Identify recently modified entries

ls -lt /tmp

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -la /tmp
total 12
drwxrwxrwt 11 root      root       300 Sep  6 10:41 .
drwxr-xr-x 19 root      root      4096 Aug 11 16:57 ..
-rw-------  1 praneetha praneetha    0 Sep  6 08:33 config-err-1e0hDv
drwxrwxrwt  2 root      root        40 Sep  6 08:33 .font-unix
drwxrwxrwt  2 root      root        60 Sep  6 08:33 .ICE-unix
srw-rw-rw-  1 root      root         0 Sep  6 08:33 .iprt-localipc-DRMIpcServer
drwx------  3 root      root        60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-colord.service-Ta5oZZ
drwx------  3 root      root        60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-ModemManager.service-UD22QU
drwx------  3 root      root        60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-polkit.service-gG5bzL
drwx------  3 root      root        60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-systemd-logind.service-yXFJk1
drwx------  3 root      root        60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-upower.service-B92r9o
-r--r--r--  1 root      root        11 Sep  6 08:33 .X0-lock
drwxrwxrwt  2 root      root        60 Sep  6 08:33 .X11-unix
-rw-------  1 praneetha praneetha  418 Sep  6 08:33 .xfsm-ICE-YS34U3
drwxrwxrwt  2 root      root        40 Sep  6 08:33 .XIM-unix

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -lt /tmp
total 0
drwx------ 3 root      root      60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-colord.service-Ta5oZZ
drwx------ 3 root      root      60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-upower.service-B92r9o
-rw------- 1 praneetha praneetha  0 Sep  6 08:33 config-err-1e0hDv
drwx------ 3 root      root      60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-ModemManager.service-UD22QU
drwx------ 3 root      root      60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-systemd-logind.service-yXFJk1
drwx------ 3 root      root      60 Sep  6 08:33 systemd-private-15a103d06e2848699ef3f9e29d496763-polkit.service-gG5bzL
```

Determine:

1. Whether files/directories currently exist.

Yes, several files and directories currently exist in /tmp, including config-err-1e0hDv, .font-unix, .ICE-unix, .iprt-localipc-DRMIpcServer, several systemd-private-* directories, .X0-lock, .X11-unix, .xfsm-ICE-YS34U3, and .XIM-unix.

2. Whether hidden files exist.

Yes, hidden entries exist, including .font-unix, .ICE-unix, .X0-lock, .X11-unix, .xfsm-ICE-YS34U3, and .XIM-unix.

3. Who owns some of the entries.

Most entries are owned by root:root, while config-err-1e0hDv and .xfsm-ICE-YS34U3 are owned by praneetha:praneetha.

4. Which entries were modified recently.

Most entries in /tmp were modified on **September 6 at 08:33**, including config-err-1e0hDv, .font-unix, .ICE-unix, .iprt-localipc-DRMIpcServer, the systemd-private-* directories, .X0-lock, .X11-unix, .xfsm-ICE-YS34U3, and .XIM-unix.

Answer:

> Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories such as /tmp can be relevant during a cybersecurity investigation because they may contain temporary files, scripts, logs, or other artifacts created by users or applications, which can provide useful evidence about system activity or suspicious behavior.

Exercise 14 - Explore the `/proc` Filesystem

Investigate:

/proc

```
```

Examine:

cat /proc/cpuinfo

cat /proc/meminfo

```
```

Then identify your current shell PID:

echo $$

```
```

Check whether a corresponding directory exists under `/proc`.

Commands Used

Examine CPU information

cat /proc/cpuinfo

Examine memory information

cat /proc/meminfo

Identify the current shell PID

echo $$

Check the corresponding process directory

ls -ld /proc/2132

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 140
model name      : 11th Gen Intel(R) Core(TM) i5-1155G7 @ 2.50GHz
stepping        : 2
microcode       : 0xffffffff
cpu MHz         : 2495.998
cache size      : 8192 KB
physical id     : 0
siblings        : 2
```

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ cat /proc/meminfo
MemTotal:        4011360 kB
MemFree:         1936804 kB
MemAvailable:    3017336 kB
Buffers:          169416 kB
Cached:          1090612 kB
SwapCached:            0 kB
Active:          1558248 kB
Inactive:         237656 kB
Active(anon):     556172 kB
Inactive(anon):        0 kB
Active(file):    1002076 kB
Inactive(file):   237656 kB
Unevictable:         112 kB
Mlocked:             112 kB
SwapTotal:       2186236 kB
SwapFree:        2186236 kB
Zswap:                 0 kB
Zswapped:              0 kB
```

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ echo $$
2132
```

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -ld /proc/2132
dr-xr-xr-x 9 praneetha praneetha 0 Sep  6 08:39 /proc/2132

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls /proc/2132
arch_status  cgroup      coredump_filter     exe      io                 loginuid   mountinfo   ns         oom_s
attr         clear_refs  cpu_resctrl_groups  fd       ksm_merging_pages  map_files  mounts      numa_maps  pagem
autogroup    cmdline     cwd                 fdinfo   ksm_stat           maps       mountstats  oom_adj    patch
auxv         comm        environ             gid_map  limits             mem        net         oom_score  perso
```

Answer:

1. What information does `/proc/cpuinfo` provide?

/proc/cpuinfo provides information about the system's processor, including the CPU model, processor number, CPU speed, number of cores, cache size, and supported CPU features.

2. What information does `/proc/meminfo` provide?

/proc/meminfo provides information about the system's RAM and swap memory, including total memory, free memory, available memory, cached memory, and swap space.

3. Why do directories with numbers appear inside `/proc`?

Numbered directories represent running processes. Each directory name corresponds to the Process ID (PID) of a process. For example, the current shell has PID 2132, so its process information is available under /proc/2132.

4. Is `/proc` an ordinary disk directory?

No. /proc is a virtual filesystem provided by the Linux kernel. Its contents are generated dynamically and provide information about running processes and the current state of the system rather than being stored as ordinary files on disk.

5. Why could `/proc` be useful during security investigation?

/proc can provide useful information about running processes, process IDs, memory usage, open files, command lines, environment information, and system activity. This can help investigators identify suspicious processes or understand what was running on a system during an investigation.

Exercise 15 - Explore `/dev`

/dev

```

**Examine:**

ls -l /dev/null: /dev/null is a character special device, owned by root:root, with major number 1 and minor number 3.

ls -l /dev/zero: /dev/zero is a character special device, owned by root:root, with major number 1 and minor number 5.

```

Commands Used:

Determine the file type of /dev/null

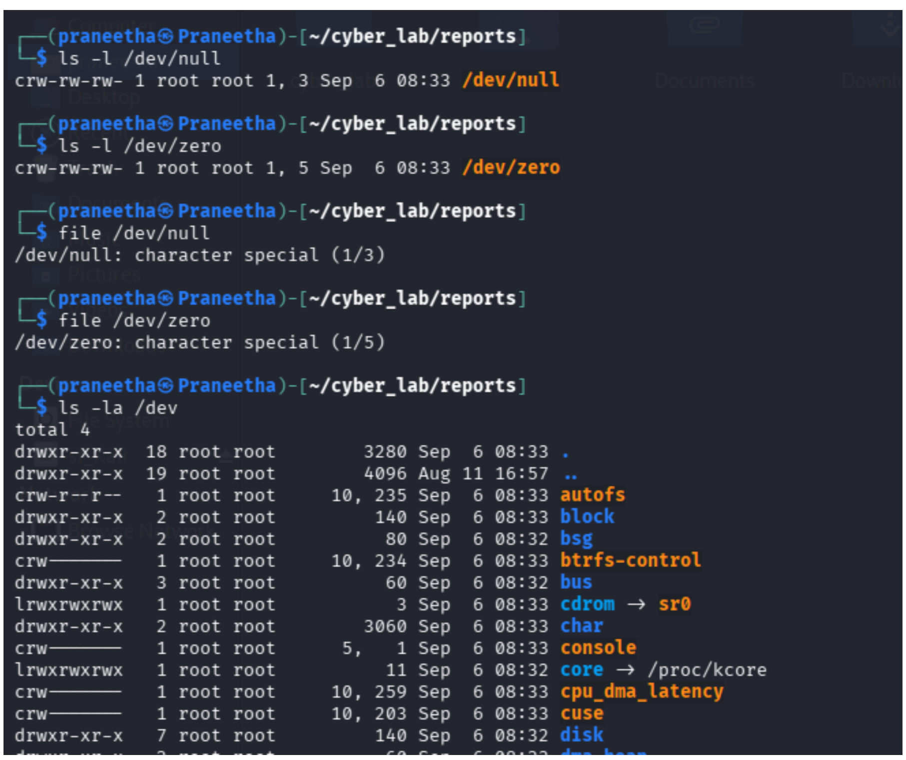
file /dev/null

Determine the file type of /dev/zero

file /dev/zero

List the contents of /dev

ls -la /dev



Answer:

1. What does `/dev` represent?

/dev represents the device filesystem in Linux. It contains special files that provide interfaces to hardware devices and virtual devices used by the operating system.

2. Are the entries ordinary user files?

No. The entries in /dev are generally device files, directories, or symbolic links, rather than ordinary user-created files. For example, /dev/null and /dev/zero are character special devices.

3. Determine the file type of `/dev/null`.

/dev/null is a character special device.

The file command produced:

/dev/null: character special (1/3)

Also, the first character c in crw-rw-rw- indicates that it is a character device.

4. Why does Linux expose devices through the filesystem?

Linux exposes devices through the filesystem so that programs can interact with hardware and system devices using standard file operations such as reading and writing. This provides a consistent way for applications and the operating system to communicate with devices.

Exercise 16 - Filesystem Space Investigation

Determine:

1. Total filesystem capacity.

2. Used space.

3. Available space.

4. Percentage of filesystem utilization.

5. Space occupied by your home directory.

6. Space occupied by the `cyber_lab` directory.

Use appropriate combinations of:

df

df -h

du

du -h

du -sh

Commands Used

Check filesystem space

df

Display filesystem space in human-readable form

df -h

Check space used by the home directory

du ~

Display home directory usage in human-readable form

du -h ~

Display total space used by the home directory

du -sh ~

Display total space used by cyber_lab

du -sh ~/cyber_lab

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ df
Filesystem     1K-blocks      Used Available Use% Mounted on
udev             1969876         0   1969876   0% /dev
tmpfs             401136      1004    400132   1% /run
/dev/sda1       38818336  18310996  18503272  50% /
tmpfs            2005680         4   2005676   1% /dev/shm
none                1024         0      1024   0% /run/credentials/systemd-journald.service
tmpfs            2005680      1956   2003724   1% /tmp
Kali           211430396 201976740   9453656  96% /media/sf_Kali
none                1024         0      1024   0% /run/credentials/getty@tty1.service
tmpfs             401136       108    401028   1% /run/user/1000

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           392M 1004K  391M   1% /run
/dev/sda1        38G   18G   18G  50% /
tmpfs           2.0G  4.0K  2.0G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.0G  2.0M  2.0G   1% /tmp
Kali            202G  193G  9.1G  96% /media/sf_Kali
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           392M  108K  392M   1% /run/user/1000

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ du ~
4       /home/praneetha/.gnupg/private-keys-v1.d
8       /home/praneetha/.gnupg
4       /home/praneetha/Downloads/suspicious_file_lab/backup
```

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ du -h ~
4.0K    /home/praneetha/.gnupg/private-keys-v1.d
8.0K    /home/praneetha/.gnupg
4.0K    /home/praneetha/Downloads/suspicious_file_lab/backup
4.0K    /home/praneetha/Downloads/suspicious_file_lab/quarantine
84K     /home/praneetha/Downloads/suspicious_file_lab
96K     /home/praneetha/Downloads
8.0K    /home/praneetha/.java/.userPrefs/burp
12K     /home/praneetha/.java/.userPrefs
16K     /home/praneetha/.java
4.0K    /home/praneetha/Videos
5.1M    /home/praneetha/Pictures
4.0K    /home/praneetha/.ssh/agent
```

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ du -sh ~
82M     /home/praneetha

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ du -sh ~/cyber_lab
32K     /home/praneetha/cyber_lab
```

1. Total filesystem capacity

Answer: The main root filesystem /dev/sda1 has a total capacity of **38 GB**.

2. Used space

Answer: 18 GB of the root filesystem is currently used.

3. Available space

Answer: 18 GB is available on the root filesystem.

4. Percentage of filesystem utilization

Answer: The root filesystem is **50% utilized**.

5. Space occupied by the home directory

Answer: The home directory /home/praneetha occupies **82 MB**.

6. Space occupied by the cyber_lab directory

Answer: The cyber_lab directory occupies **32 KB**.

Answer:

> Explain the difference between `df` and `du`.

df reports the overall disk space available and used on a filesystem, while du reports the space occupied by individual directories and files.

For example, df -h showed that /dev/sda1 has 38 GB total, 18 GB used, and 18 GB available, while du -sh ~ showed that the user's home directory occupies 82 MB.

Exercise 17 - Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported.

You are initially asked to examine only the filesystem.

Identify which of the following locations you would investigate and explain why:

/home

/etc

/var/log

/tmp

/proc

/dev

/root

For each selected location, state what type of security-relevant information you would expect to find.

Location	**Investigation**	**Security-relevant information**
/home	Yes	User files, shell histories, scripts, and downloaded files.
/etc	Yes	Authentication, user accounts, system configuration, and scheduled tasks.
/var/log	Yes	System, login, authentication, service, and error logs showing activity.
/tmp	Yes	Temporary files, scripts, sockets, and possible suspicious artifacts.
/proc	Yes	Running processes and live CPU, memory, and system information.
/dev	Yes	Device files and interfaces; unusual device access may be relevant.
/root	Yes	Root user files, histories, scripts, and evidence of privileged activity.

Submit the following:

1. Commands used for each exercise.

Commands Used

/home

ls -la /home

find /home -maxdepth 2 -type f -print

/etc

ls -la /etc

find /etc -maxdepth 1 -type f -print

/var/log

ls -la /var/log

find /var/log -maxdepth 1 -type f -print

/tmp

ls -la /tmp

find /tmp -maxdepth 1 -type f -print

/proc

ls -la /proc

cat /proc/cpuinfo

cat /proc/meminfo

/dev

ls -la /dev

ls -l /dev/null

ls -l /dev/zero

/root

ls -la /root

2. Relevant output or screenshots.

```
┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ ls -la /home
total 24
drwxr-xr-x  6 root      root      4096 Aug 31 21:41 .
drwxr-xr-x 19 root      root      4096 Aug 11 16:57 ..
drwx------  5 alice     alice     4096 Sep  2 22:05 alice
drwx------  5 bob       bob       4096 Aug 31 21:41 bob
drwx------  5 charlie   charlie   4096 Sep  2 22:32 charlie
drwx-----x 19 praneetha praneetha 4096 Sep  6 09:55 praneetha

┌──(praneetha㉿Praneetha)-[~/cyber_lab/reports]
└─$ find /home -maxdepth 2 -type f -print
find: '/home/charlie': Permission denied
find: '/home/alice': Permission denied
/home/praneetha/.gitconfig
/home/praneetha/.vboxclient-draganddrop-tty7-service.pid
/home/praneetha/.bashrc.original
/home/praneetha/.vboxclient-hostversion-tty7-control.pid
/home/praneetha/.profile
/home/praneetha/.zprofile
/home/praneetha/.vboxclient-seamless-tty7-control.pid
/home/praneetha/.vboxclient-clipboard-tty7-service.pid
/home/praneetha/.bash_history
/home/praneetha/.face
/home/praneetha/.zshrc
/home/praneetha/.vboxclient-vmsvga-session-tty7-service.pid
/home/praneetha/.vboxclient-draganddrop-tty7-control.pid
/home/praneetha/.vboxclient-clipboard-tty7-control.pid
/home/praneetha/.bash_logout
/home/praneetha/.zsh_history
/home/praneetha/.vboxclient-vmsvga-session-tty7-control.pid
/home/praneetha/.sudo_as_admin_successful
/home/praneetha/.xsession-errors.old
/home/praneetha/.vboxclient-seamless-tty7-service.pid
/home/praneetha/.Xauthority
/home/praneetha/.ICEauthority
/home/praneetha/.xsession-errors
/home/praneetha/.bashrc
/home/praneetha/.dmrc
find: '/home/bob': Permission denied
```

3. Observation tables.

Location	Observation
/home	Multiple user directories and hidden history/configuration files were found.
/etc	Files such as passwd, shadow, sudoers, crontab, hosts, and hostname were present.
/var/log	Logs including btmp, wtmp, boot.log, dpkg.log, and service logs were found.
/tmp	Temporary files, X11 files, sockets, and systemd private directories were present.
/proc	Numerous process-ID directories and system information files were present.
/dev	Device entries such as /dev/null, /dev/zero, /dev/sda, and its partitions were present.
/root	Access was denied to the current user, indicating restricted permissions.

4. Disk → Partition → Filesystem → Mount Point diagram.

```
/dev/sda (40 GB Disk)
│
├── /dev/sda1 (37.9 GB Partition)
│     └── ext4 Filesystem
│           └── / (Root Mount Point)
│
├── /dev/sda2 (1 KB Partition)
│     └── No filesystem shown
│
└── /dev/sda5 (2.1 GB Partition)
      └── swap Filesystem
            └── [SWAP]
```

5. Answers to analysis questions.

1. Which locations should be investigated?

All seven locations should be investigated: /home, /etc, /var/log, /tmp, /proc, /dev, and /root, because each provides a different type of security-relevant information.

2. Why is /home important?

It contains user data, hidden files, shell histories, scripts, and other files that can reveal user activity.

3. Why is /etc important?

It contains system and security configuration files that can reveal account, authentication, privilege, network, and scheduled-task information.

4. Why is /var/log important?

It contains logs that can provide evidence of previous system, login, application, service, and boot activity.

5. Why is /tmp important?

Temporary files and artifacts may reveal scripts, applications, or suspicious activity that occurred on the system.

6. Why is /proc important?

It provides information about running processes and the current state of system resources such as CPU and memory.

7. Why is /dev important?

It contains device files that provide access to disks, terminals, and other system devices.

8. Why is /root important?

It may contain privileged user files and evidence of administrative activity. Although access was denied during the investigation, it remains an important location to examine with appropriate privileges.